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Important Quantum Gates for Quantum Algorithms of Travelling Salesman Problem

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***Abstract*—This paper proposes $B(\frac{1}{k})$ gates as important gates to generate W state for the quantum approximate optimization algorithm (QAOA) to tackle the popular optimization problem known as the traveling salesman problem (TSP). We also provide a simple proof for the $B(\frac{1}{k})$ gates followed by the corresponding quantum circuit. We present the Ising Hamiltonian tailored specifically for the TSP, including its associated constraints. We perform a performance comparison for QAOA with the conventional state initialization method involving the Hadamard gate and QAOA with state initialization using the W state. We found that the performance of QAOA is significantly influenced by two key factors of (i) the number of p layers and (ii) the chosen state initialization method. Our findings highlight that the generation and integration of W state initialization results in a substantial reduction in the search space for QAOA, consequently reducing the required number of p layers for the search process.**

***Keywords*—quantum approximate optimization algorithm, travelling salesman problem, state initialization, Hadamard gate, W state.**

I. Introduction

Travelling salesman problem (TSP) is a classical optimization problem in the field of computer science. Essentially, the TSP seeks to identify a route that visits N number of cities exactly once and returning to city of origin with the least amount of resources used [1]. Each city is connected by an edge and can be expressed with cost or weight of travel $w_{i,j}$. For a complete and symmetrical N-city ($w_{i,j} = w_{j,i}$) TSP, there are $\frac{(N-1)!}{2}$ possible routes. As the number of cities increases, the complexity of the problem grows exponentially. For this reason, TSP is classified as an NP-hard problem [2], [3]. A quick and efficient way of TSP solution can accelerate the development of, for example, intelligent transportation system (ITS), offering substantial benefits to the transportation and logistics industry [4].

The rapid development of quantum computing has paved the way for new algorithms aimed at addressing the TSP. The impact of quantum computing is also noticeable in the further development of error correction codes [5], [6]. In the noisy intermediate-scale quantum (NISQ) era, several candidates of hybrid quantum algorithms, such as variational quantum eigensolver (VQE) and quantum approximate optimization algorithm (QAOA) [7], [8] have been developed to solve TSP and other complex NP-hard optimization problems. Hybrid quantum algorithms offer distinct advantages in the NISQ era since hybrid quantum algorithms take leverage of classical computer, making the algorithm more robust to errors. This paper specifically focuses on QAOA and efficient initialization techniques used to improve the performance of QAOA.

QAOA has been heavily studied to solve another optimization problem, e.g., the Max-cut problem for its simplicity and its connection to the Ising Model [9]. QAOA is chosen as a way to solve TSP because, its performance has been proven to achieve the approximation ratio of at least 0.6924 on 3-regular graphs for p = 1, where p is the depth of the quantum circuit [10]. This track record makes QAOA as a promising candidate for solving various optimization problems including TSP.

To fully encode the solution to an N-city TSP, we need N^2 qubits. The solution to a TSP can be expressed as a series of bits such that it represents a specific Hamiltonian cycle. An example of a series of bitstream x representing a solution to

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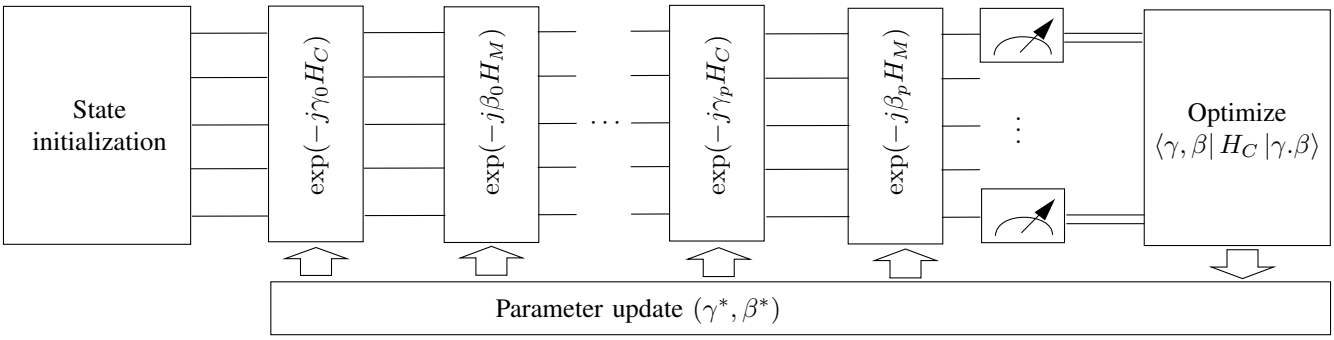

Fig. 1. The general workflow of QAOA.

a typical 3-city TSP problem can be written as

$$x = (1 \to 2 \to 3) \equiv \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \equiv 100\ 010\ 001, \quad (1)$$

where the first part of the bitstream is from the first row of the matrix followed by the second and the third rows. City-i is visited at time t if $x_{i,t} = 1$, otherwise the city is not visited in the solution. This method makes it challenging to solve TSP, especially when dealing with a large number of destinations since N^2 qubits are needed. On the other hand, we only need exactly bit '1' every N bits in different positions, otherwise the solution for the TSP would violate the constraints of TSP. In this paper, we propose a new way of state initialization in QAOA by using the W state [11], a commonly used concept in quantum communications, because the properties of W state satisfy our requirements. We summarise the contribution of this paper as follows: (i) a $B(\frac{1}{k})$ gate to generate W state and (ii) utilising the W state as an initial state of QAOA for solving the practical TSP.

II. Method and The Proposed Gates

A. Quantum Approximate Optimization Algorithm (QAOA)

QAOA is closely related with the adiabatic quantum computing (AQC), which operates based on the adiabatic principle [12]. The general structure of QAOA consists of two Hamiltonian operators, i.e., the cost Hamiltonian and the mixer Hamiltonian. The cost Hamiltonian layer can be written as a unitary operator $U(H_C, \gamma)$, which depends on parameter γ as

$$U(H_C, \gamma) = e^{-i\gamma H_C}, \quad (2)$$

where H_C is the cost Hamiltonian for the specific problem. The mixer Hamiltonian layer can be written as a unitary operator $U(H_M, \beta)$, which depends on parameter β as

$$U(H_M, \beta) = e^{-i\beta H_M}, \quad (3)$$

where H_M is the mixer Hamiltonian for the specific problem. The number of layers applied can be defined as an integer p. For $p \geq 1$, the angle dependent quantum state produced by the algorithm can be written as an alternating operation between (2) and (3) on the initial state as shown in Fig. 1[1] and can be mathematically written as

$$\begin{aligned} |\gamma, \beta\rangle &= U(H_M, \beta_p)U(H_C, \gamma_p)\cdots \\ &\quad U(H_M, \beta_0)U(H_C, \gamma_0)\,|\psi_0\rangle\,. \end{aligned} \quad (4)$$

The goal of this algorithm is to enhance the expectation value of H_C, based on the objective function, such that the approximation ratio α is closer to 1. The expectation value of H_C can be written as $\langle f\rangle$, while the approximation ratio α can be defined as

$$\langle f\rangle = \langle\gamma, \beta|\, H_C\, |\gamma, \beta\rangle\,, \quad (5)$$

$$\alpha = \frac{\langle\gamma, \beta|\, H_C\, |\gamma, \beta\rangle}{H_{C_{optimal}}} \geq 1, \quad (6)$$

respectively. This goal can be achieved by searching γ and β that satisfy minimal $\langle f\rangle$. By combining this quantum algorithm and classical search routine, we can find the best γ and β.

B. State Initialization

The standard technique to prepare initial state is by applying a series of Hadamard gates to all qubits that put all qubit states in an equal superposition. For an N-city TSP, the initial state can be defined as

$$|\psi_0\rangle = H^{\otimes N^2}\,|0\rangle = |+\rangle^{\otimes N^2} = \frac{1}{\sqrt{2^{N^2}}}\sum_{x=0}^{2^{N^2}-1}|x\rangle\,. \quad (7)$$

This simple initialization technique comes with its own limitations. By putting all qubits in a uniform superposition, invalid solutions that violate the penalties of TSP will also be in an equal superposition causing almost equal probability for both valid and invalid solutions. For this reason, state initialization plays a significant role in QAOA, as it can increase or reduce the number of solutions we are dealing with.

One of the initialization states we have chosen in this paper is the W state, which is mathematically written as

$$\begin{aligned} |W\rangle &= \frac{1}{\sqrt{N}}(|100\cdots0\rangle + |010\cdots0\rangle \\ &\quad + \cdots + |000\cdots1\rangle). \end{aligned} \quad (8)$$

[1]Measurement is not discussed in this paper.

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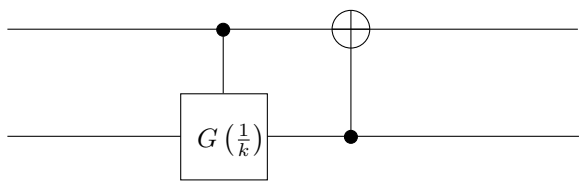

Fig. 2. The structure of the proposed $B\left(\frac{1}{k}\right)$ gate as defined in (10).

By applying the W state, we can decrease the search space of QAOA and eliminate several invalid solutions. The construction of W state depends on the gate $B(p)$ proposed in [13]. To avoid confusion between QAOA layer p in this paper and parameter p in $B(p)$ gate in [13], we substitute p in $B(p)$ to be $\frac{1}{k}$ in $B(\frac{1}{k})$, $k = \{2, 3, \cdots, N\}$. For an N-qubit system, the value of $\frac{1}{k}$ ranges from $\frac{1}{N}$ until $\frac{1}{2}$. In this paper, we propose the construction of W state from $B\left(\frac{1}{k}\right)$ consisting of several controlled-$G\left(\frac{1}{k}\right)$ gates followed by an upside down controlled-NOT (CNOT) of

$$\bar{C}_n = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{9}$$

as shown in Fig. 2 such that

$$B\left(\frac{1}{k}\right) = \bar{C}_n \cdot C_{G\left(\frac{1}{k}\right)} \tag{10}$$

with $C_{G(\frac{1}{k})}$ being controlled-$G(\frac{1}{k})$ defined as

$$C_{G\left(\frac{1}{k}\right)} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \sqrt{\frac{1}{k}} & -\sqrt{1-\frac{1}{k}} \\ 0 & 0 & \sqrt{1-\frac{1}{k}} & \sqrt{\frac{1}{k}}. \end{bmatrix}. \tag{11}$$

Corollary II.1. *A W state of N qubits can be constructed from $(N-1)$ $B(\frac{1}{k})$ gates.*

Proof. For $N = 3$, we can construct a W_3 state using several $C_{G(\frac{1}{k})}$ followed by $\bar{C}_n$ with descending k as shown in Fig. 3 as

$$\begin{aligned} |W_3\rangle &= \left[I \otimes \left[C_n \cdot C_{G(\frac{1}{2})}\right]\right] \left[\left[C_n \cdot C_{G(\frac{1}{3})}\right] \otimes I\right] |100\rangle \\ &= \left[I \otimes \left[C_n \cdot C_{G(\frac{1}{2})}\right]\right] \left[\sqrt{\frac{2}{3}}\,|010\rangle + \sqrt{\frac{1}{3}}\,|100\rangle\right] \\ &= \sqrt{\frac{1}{3}}\,|001\rangle + \sqrt{\frac{1}{3}}\,|010\rangle + \sqrt{\frac{1}{3}}\,|100\rangle\,. \end{aligned} \tag{12}$$

For general N, the W state can be constructed from $B(\frac{1}{k})$ as

$$\begin{aligned} |W\rangle &= B\left(\frac{1}{2}\right) B\left(\frac{1}{3}\right) \cdots \\ &\quad B\left(\frac{1}{k-1}\right) B\left(\frac{1}{k}\right) |100\cdots00\rangle \end{aligned} \tag{13}$$

of which the quantum circuit is shown in Fig. 4. The proof is completed. ■

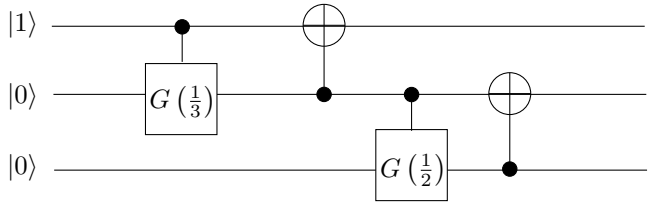

Fig. 3. The proposed structure of W_3 state using $B(\frac{1}{k})$ with $k = \{2, 3\}$.

It is quite clear to construct a W state. We require $(N-1)$ operations of $B\left(\frac{1}{k}\right)$ and $2\times(N-1)$ total operations of $C_{G\left(\frac{1}{k}\right)}$ and upside down CNOT. This state initialization technique can immensely impact the performance of QAOA by reducing the number of solution candidates.

Various other ways of state preparation in QAOA have been introduced to avoid invalid solutions such as constructing an oracle function discussed in [14], [15]. The oracle is commonly used for unstructured database search, for example, Grover's search algorithm [16]. The oracle function validates and checks if a solution meets the corresponding problem constraints such that the algorithm starts with only feasible solutions. Other than modifying the initial state, the performance of QAOA can also be improved by warm-starting the initial parameter using parameter fixing strategy [17]. The right parameter fixing strategy can reduce the amount of layers used in the algorithm and avoid barren plateaus during optimization process.

C. Cost Hamiltonian

We express the cost function of TSP as [18]

$$C_{cost} = \sum_{0\le i,j<N} w_{i,j} \sum_{t=0}^{N-1} x_{i,t} x_{j,t+1}. \tag{14}$$

By finding the value of x that minimizes (14), we can find the solution of a TSP with constraints

$$\sum_{i=0}^{N-1} x_{i,t} = 1, \quad \forall t \in \{0, \ldots, N-1\}, \tag{15}$$

$$\sum_{t=0}^{N-1} x_{i,t} = 1, \quad \forall i \in \{0, \ldots, N-1\}. \tag{16}$$

Equation (15) prevents the solution to simultaneously be in several cities, while (16) prevents the solution to visit the same city more than once. The complete cost function of TSP can be written as the summation of (14), (15), and (16) as

$$\begin{aligned} C_{TSP} &= \sum_{0\le i,j<n} w_{i,j} \sum_{t=0}^{n-1} x_{i,t} x_{j,t+1} \\ &\quad + A \sum_{t=0}^{n-1} \left[1 - \sum_{i=0}^{n-1} x_{i,t}\right]^2 \\ &\quad + A \sum_{i=0}^{n-1} \left[1 - \sum_{t=0}^{n-1} x_{i,t}\right]^2, \end{aligned} \tag{17}$$

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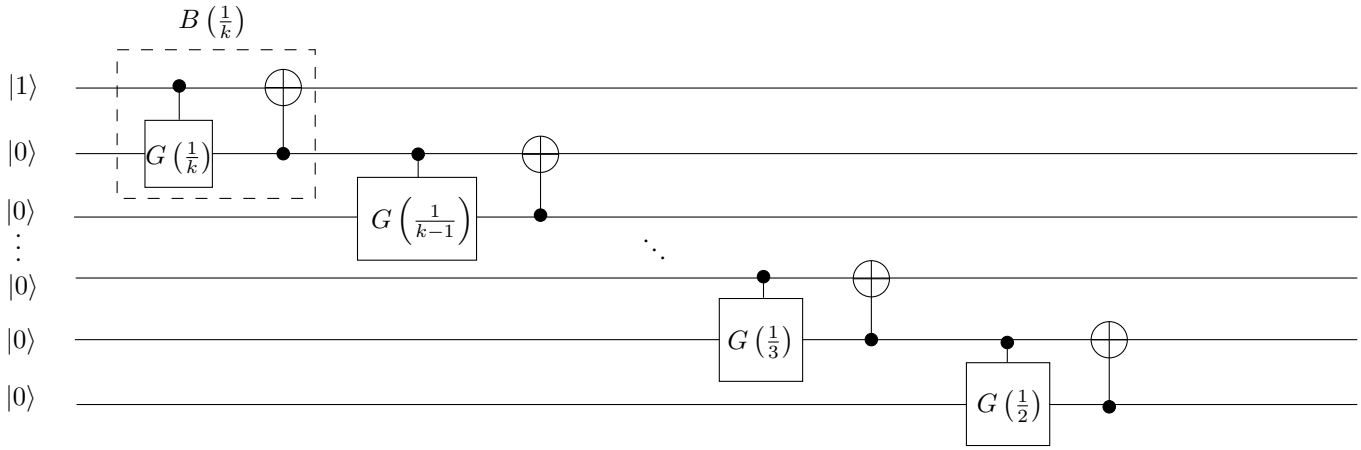

Fig. 4. The general structure of the W state for state initialization.

where A is the penalty term. The value of A should be positive and ideally larger than the largest $w_{i,j}$. To construct the complete Ising Hamiltonian for the complete cost function, we can substitute $x_{i,j}$ using spin variable $\sigma^z_{i,j}$. The binary mapping function can be written as

$$x_{i,t} = \frac{I_{i,t} - \sigma^z_{i,j}}{2}, \tag{18}$$

where $I_{i,t}$ and $\sigma^z_{i,j}$ can be expressed as

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad \sigma^z_{i,j} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{19}$$

The final cost Hamiltonian is obtained by substituting (18) to (17). The complete cost Hamiltonian of TSP is composed of the Pauli-Z. The cost Hamiltonian of TSP can be written as

$$\begin{aligned} H_C = H_{TSP} &= H_{cost} + H_{penalty}, \\ &= \sum_i c_i \sigma^z_i + \sum_{i,j} c_{i,j} \sigma^z_i \sigma^z_j. \end{aligned} \tag{20}$$

We can combine (20) and (2) to obtain the complete cost Hamiltonian layer for TSP. σ^z_i is a Pauli operator acting on the i-th qubit, while σ^z_i σ^z_j can be realized using a rotation RZ gate between two CNOT gates, otherwise known as RZZ gate with the i-th qubit as control and the j-th qubit as target. The cost Hamiltonian can simply be expressed in a quantum circuit using RZ and RZZ gates, which are mathematically written as

$$RZ(\gamma) = \begin{bmatrix} e^{-j\frac{\gamma}{2}} & 0 \\ 0 & e^{j\frac{\gamma}{2}} \end{bmatrix}, \tag{21}$$

$$RZZ(\gamma) = \begin{bmatrix} e^{j\frac{\gamma}{2}} & 0 & 0 & 0 \\ 0 & e^{-j\frac{\gamma}{2}} & 0 & 0 \\ 0 & 0 & e^{-j\frac{\gamma}{2}} & 0 \\ 0 & 0 & 0 & e^{j\frac{\gamma}{2}} \end{bmatrix}, \tag{22}$$

respectively.

D. Mixer Hamiltonian

The mixer Hamiltonian maps the algorithm to achieve feasible solutions. The mixer Hamiltonian also makes sure the algorithm favors feasible search spaces over infeasible solutions. There has been several mixer Hamiltonian strategies, for example, as proposed in [19]. In this paper, we are going to focus only on the traditional QAOA mixer Hamiltonian, which can be defined as

$$H_M = \sum_i \sigma^x_i, \tag{23}$$

where σ^x_i can be expressed as

$$\sigma^x_i = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{24}$$

We can construct the QAOA mixer layer by combining (23) and (3). The implementation of mixer Hamiltonian on a quantum circuit is simply by applying a series of rotation RX gates, which can be mathematically written as

$$RX(\beta) = \begin{bmatrix} \cos\left(\frac{\beta}{2}\right) & -j\sin\left(\frac{\beta}{2}\right) \\ -j\sin\left(\frac{\beta}{2}\right) & \cos\left(\frac{\beta}{2}\right) \end{bmatrix}. \tag{25}$$

E. Measurement and Optimization

Since the building blocks for QAOA have been explained in previous subsections, this subsection focuses on the final step of QAOA, i.e., measurements and classical optimization. We measure the expectation value in (5) after performing alternating operations of (2) and (3) on the initial state (7). The classical optimization process varies the values of (γ, β) p times until the condition of α in (6) equals to one or until the maximum amount of iterations is reached. Lower value of α in (6) indicates a lower expectation of the solution state for TSP. Ideally, larger values of repetition p improves the performance of QAOA. As mentioned in [20], several classical optimizers such as constrained optimization by linear approximation (COBYLA) algorithm and simultaneous perturbation stochastic approximation (SPSA) algorithm are proven

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Fig. 5. Map of Telkom University showing the stations of TUC.

to perform better at higher values of p compared to other classical optimizers.

III. Simulation Results and Analysis

We evaluate a case of TSP using the proposed W state and brute force method for fairness. We try to solve a simple case of TSP in real life, which is finding the best route for Telkom University's private mode of transportation, called Telkom University car (TUC). TUC has several stations across the campus. Some of the stations are located at Telkom University Landmark Tower (TULT) marked as a, the dormitory parking spot as b, the public lecture building as c, and the multipurpose hall as d. The full map of Telkom University is shown in Fig. 5 with map pins indicating the station locations.

The approximate distances between each station are expressed in meters and is written in form of matrix such as

$$C_{\text{TUC}} = \begin{bmatrix} 0 & 400 & 700 & 1600 \\ 400 & 0 & 300 & 1300 \\ 700 & 300 & 0 & 850 \\ 1600 & 1300 & 850 & 0 \end{bmatrix}. \tag{26}$$

The case mentioned is rather simple but can be problematic for the algorithm due to N^2 number of qubits used, especially for a TSP with large number of cities. The case is solved using traditional state initialization expressed in (7) and W state initialization expressed in (8).

This case study requires $N^2 = 4^2 = 16$ qubits to solve. We also compare and show the performance of QAOA using different initialization techniques mentioned in previous section to further show the advantages of using W state initialization. Brute force method provides the best route for TUC of $(a \rightarrow b \rightarrow c \rightarrow d)$ with a total distance of 3150 meters. The solution can be expressed using bitstream 1000 0100 0010 0001.

Fig. 6 shows the probability distribution for the TSP case without W state showing a poor performance for $p = 1$ that does not give a valid and good result, although they look like valid with small number of peaks. This poor performance is expected since there is no particular state preparation strategy implemented on this algorithm. We try to solve the same problem using W state preparation with the better result being shown in Fig. 7, where the best result is the state of $|0010\ 0001\ 1000\ 0100\rangle$. This state corresponds to the route $(c \rightarrow d \rightarrow a \rightarrow b)$, which is equivalent to route $(a \rightarrow b \rightarrow c \rightarrow d)$ of the brute force search indicating the validity of QAOA using W state.

The W state proves to be a simple yet effective way of improving the performance of QAOA for solving TSP even with small number of p. The proposed QAOA successfully demonstrates in solving real-world TSP instances with up to four cities. Compared to the QAOA with traditional Hadamard initialization, the proposed QAOA yields a more plausible and valid solution. This improvement is attributed to the W state's unique property of assigning precisely a single bit '1' for every N bits. This technique completely eliminates the probability of solutions that occupy several cities simultaneously. The only penalty left to deal with is the prohibition of revisiting the same city. By utilising the W state as the initial state, we can reduce the initial complexity of the proposed algorithm from 2^N to N.

We also find that the increasing TSP size beyond four cities may escalate the complexity of the proposed algorithm and the execution time. The number of qubits can be further reduced by fixing a starting point to eliminate several permutations of the same solutions from N^2 into $(N-1)^2$ [18]. Another way of reducing the number of qubits is by using an oracle-based function to eliminate incorrect and invalid solutions to decrease the search space and improve the performance of QAOA. This method can decrease the number of qubits from N^2 into $\frac{N(N-1)}{2}$ [14], which is a significant reduction.

IV. Conclusion

We have proposed $B(\frac{1}{k})$ gates as important gates for QAOA for TSP followed by the proof and the corresponding quantum circuit. We have also implemented the proposed $B(\frac{1}{k})$ gates to generate W state as an initialization state in QAOA for solving the famous NP-hard TSP problem. We have demonstrated and found that a simple modification on the state initialization in QAOA using W state improves the performance of the proposed algorithm. Additionally, we have also provided a general framework to construct the W state, which is applicable on various quantum systems. The QAOA is also proven to be reliable using only a small number of p layers. However, the limitation of QAOA lies on the number of qubits, which is infeasible for large problems. Both cases in this paper require N^2 qubits, which is expensive for solving a relatively small size TSP. Overcoming this challenge has a potential to elevate QAOA into a robust algorithm suitable for efficiently tackling a broad spectrum of real-world problems.

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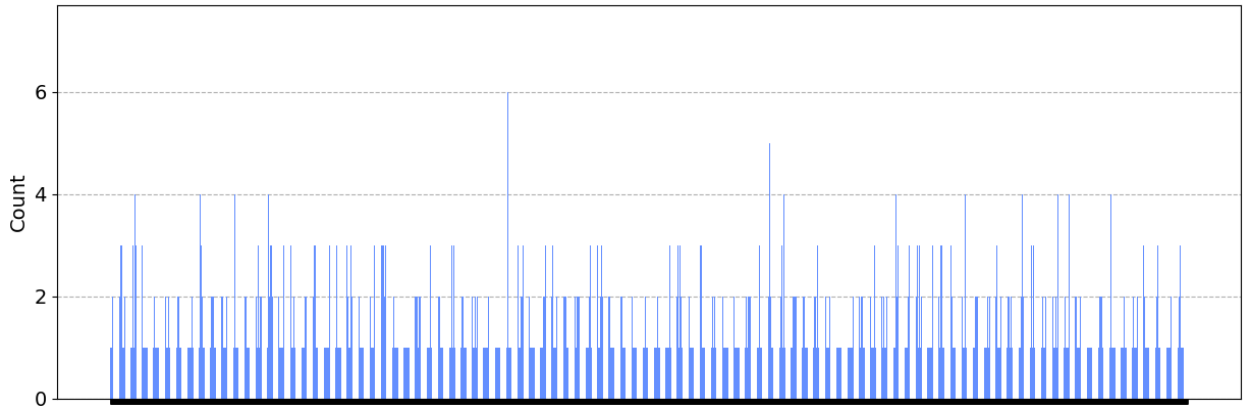

Fig. 6. Probability distribution of QAOA solution states for TUC route using $p = 1$ without W_4 state initialization.

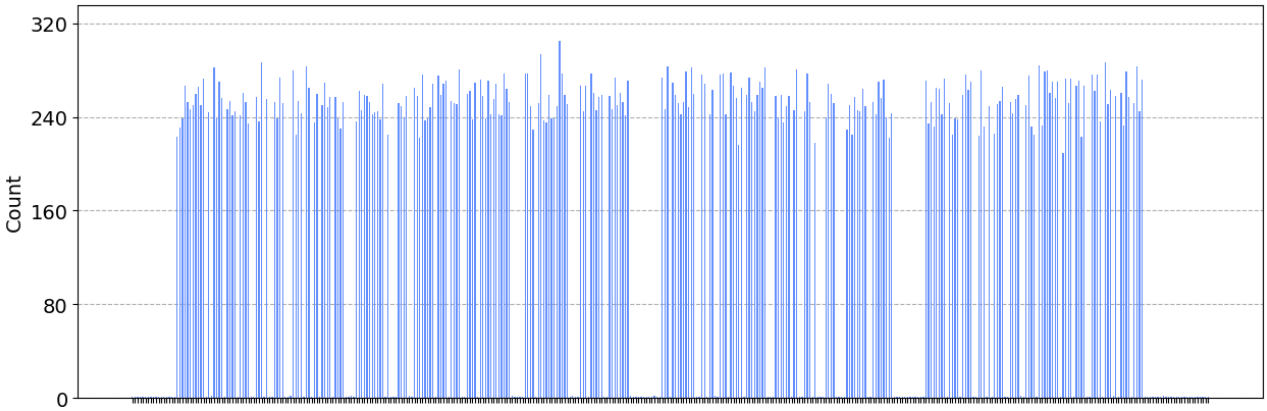

Fig. 7. Probability distribution of QAOA solution states for TUC route using $p = 1$ with W_4 state initialization.

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